DATA-DRIVEN OPERATIONAL COST CONTROL AND MENU OPTIMIZATION IN HOTEL F&B USING SQL AND MENU ENGINEERING ANALYTICS

A CASE STUDY PORTFOLIO FOR FOOD & BEVERAGE OPERATIONS ANALYSIS 2026

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CHAPTER 1

INTRODUCTION

1.1 Background of Study

The contemporary global hospitality sector operates within an increasingly volatile economic landscape characterized by fluctuating supply chains, rising raw material inflation, and highly dynamic consumer behaviors. Within a hotel ecosystem, the Food and Beverage (F&B) department serves a vital dual purpose: it acts as a primary engine for top-line revenue generation while simultaneously accounting for one of the largest segments of operational expenditure. Unlike fixed hotel overhead costs, such as property maintenance, utility fees, or administrative salaries, F&B financial metrics fluctuate dynamically on a daily basis. These fluctuations are directly influenced by hotel guest occupancy rates, banquet bookings, restaurant traffic, menu pricing structures, and raw ingredient utilization protocols.

To maintain baseline profitability, hotel operators have traditionally relied on conventional procurement-focused cost control mechanisms. These legacy methodologies emphasize bulk purchasing discounts, volume contracts, and rigorous vendor negotiations designed to reduce the initial unit cost of raw materials. While highly effective at the point of invoice, this legacy approach completely ignores the micro-inefficiencies that occur post-procurement within the active kitchen production environment. Inefficiencies such as excessive portion sizes, poor batch forecasting, and a lack of analytical visibility into historical sales patterns lead to severe cumulative losses (Dittmer & Keefe,2009). In modern hospitality analytics, achieving sustainable cost control requires shifting away from manual, backward-

looking ledger entries. Instead, operators must implement an integrated, data-driven framework that actively links raw inventory inputs, kitchen wastage variables, and actual sales volumes to clear strategic adjustments in menu design.

This study presents a comprehensive operational portfolio showcasing the development of an automated, data-driven cost control system within a simulated hotel F&B framework. By migrating raw operational logs from flat spreadsheets into a Relational Database Management System (RDBMS) utilizing Structured Query Language (SQL), this project establishes a scalable data pipeline (Nyheim & Connolly, 2014). The crest analytical engine applies to the classic principles of the Kasavana & Smith Menu Engineering Matrix. This matrix categorizes hotel menu offerings into performance-based quadrants by simultaneously calculating popularity metrics and actual contribution margins (Kasavana & Smith, 1982). Through this technological integration, the research demonstrates how raw operational metadata can be systematically converted into high-impact management decisions. This workflow optimizes financial yields while directly curbing food waste volumes.

1.2 Problem Statement

Despite the clear availability of high-volume transaction logs from Point-of-Sale (POS) systems, modern hotel F&B departments frequently suffer from severe profit leakage. This chronic financial underperformance can be traced directly to two fundamental, structural operational flaws:

1. Absence of Predictive Production Forecasting and Waste Isolation:

Traditional culinary production setups inside large hotel kitchens routinely default to over-production to avoid service delays or stockouts during peak operational hours. Because raw materials are tracked globally at the end of the month rather than programmatically isolated per menu item, specific sources of volume leakage remain completely hidden. In this case study's operational environment, this baseline flaw is demonstrated by an accumulated raw waste cost of RM5,143.30 drawn against a total production investment of RM27,646.00. This represents an immediate, unrecovered 18.6% deduction from the operational gross margin (Papargyropoulou et al., 2014). Without a system capable of isolating which specific menu items drive this variance, management is forced to absorb these losses blindly as an inevitable cost of doing business.

2. Static Pricing Structures and Blind Menu Engineering Classifications:

The second structural failure lies in the static nature of menu management. Most hotels set menu prices using a flat, unadjusted markup over perceived ingredient costs, failing to account for how popularity trends and variable wastage metrics impact actual profitability. Without an automated framework to continuously cross-reference sales volumes against real-time food cost percentages, the operation remains strategically blind. Management cannot systematically identify "Plowhorses"—high-volume items that drive foot traffic, but aggressively erode margins due to high unit costs or massive preparation waste. Conversely, they remain blind to "Puzzles" items that feature excellent margin structures but fail to sell due to poor visibility, marketing, or pricing positioning (Kasavana & Smith, 1982). The analytical dataset for this facility reveals that a massive 52% of the total menu offerings reside within the 'Puzzle' quadrant. In an environment where more than half of the menu generates high margin potential but suffers from low market velocity, failing to implement active, automated item-level tracking directly limits the hotels long-term profitability.

1.3 Research Objectives

To directly resolve the structural inefficiencies identified in the problem statement, this portfolio project fulfills the following core research objectives:

1. To isolate operational variances across hotel menu categories by measuring the precise correlation between portion volume preparation and itemized capital waste.

2. To engineer a production-ready SQL database schema and automated query structure that eliminates manual evaluation by programmatically classifying multi-category menu items into the four quadrants of the Menu Engineering Matrix.

3. To build an executive data visualization dashboard that aggregates complex relational database outputs into immediate visual indicators, enabling senior hotel leadership to execute data-driven menu adjustments.

1.4 Significance of the Study

The findings and framework developed in this study provide dual-layered value, bridging theoretical hospitality data analytics with practical enterprise operations. Firstly, for Hotel Executive Management and Financial Controllers, this research establishes an objective, audit-ready validation protocol. Instead of managing F&B departments through defensive, reactive cost-cutting (such as down-sizing staff or reducing ingredient quality), leaders can now deploy offensive, data-driven margin engineering. This protects hotel profitability while preserving the guest experience.

Secondly, for Kitchen Operations and Culinary Director Teams, this study shifts inventory management away from collective kitchen guesswork and towards automated portion control. By isolating specific item leakages, the culinary team can optimize procurement schedules and implement targeted ingredient cross-utilization. This directly minimizes the property's carbon footprint and environmental waste index (Papargyropoulou et al., 2014). Finally, from a technical perspective, this project serves as a highly scalable architecture blueprint for data analysts in the hospitality domain, demonstrating how standard relational database scripting (SQL) can seamlessly modernise legacy spreadsheet operations (Nyheim & Connolly, 2014).

1.5 Scope of the Study

The analytical boundaries of this study are restricted to a dataset tracking 35 distinct menu items designed for a full-service hotel dining facility. These items represent a balanced operational mix split across four core culinary categories: Main Courses, Appetizers, Desserts, and Beverages. Strategically, the scope of this project is explicitly built around showcasing a continuous three-tiered data engineering pipeline:

* Phase 1: Structured ingestion, validation, and data engineering of raw kitchen yields using Microsoft Excel.

* Phase 2: Relational database architecture, data manipulation, and threshold-driven matrix classification executed via MySQL queries.

* Phase 3: High-level visual synthesis and KPI tracking utilizing an advanced analytics executive dashboard.

This pipeline isolates operational performance down to individual product metrics, ensuring the technical framework can be easily scaled to larger multi-outlet hotel systems.

CHAPTER 2

LITERATURE REVIEW

2.1 Food and Beverage Cost Control and Food Waste Management

Controlling variable expenses in hotel Food and Beverage (F&B) operations represents one of the most persistent operational hurdles in hospitality management. According to industry standards, food cost differs significantly from fixed hotel overheads because it is highly sensitive to external fluctuations, inflation, supply chain inconsistencies and shifting consumer preferences (Dittmer & Keefe,2009). A critical concept in understanding financial leaks within a commercial kitchen is the phenomenon of "over-production". Over-production refers to the calculated or accidental gap between the volume of food portions prepared by the culinary team and the actual volume of portions ordered and paid for by guests.

In a large-scale hotel environment—where kitchens must simultaneously cater to a la carte dining rooms, room service, executive lounges, and large banquet events—culinary operations frequently fall back on producing excessive buffer portions to prevent service delays or stockouts. However, when these prepared portions go unsold, they turn into direct capital loss, categorized as food waste cost (Papargyropoulou et al., 2014). Traditional financial methodologies attempt to control this by implementing month-end global inventory audits, calculating a single, overarching food cost percentage for the entire property. While this global percentage satisfies basic corporate accounting requirements, it completely lacks the granularity required to diagnose item-level inefficiencies.

Modern hospitality research establishes that sustainable cost control is directly dependent on dynamic, itemized waste isolation (Miller et al., 2002). When an operation can trace exactly how much capital is lost to wastage per menu item, it can pinpoint operational failures—such as improper portion tracking, poor raw ingredient shelf-life management, or kitchen preparation errors. Reducing this individual item variance is the most direct method a hotel can use to stabilize its overall food cost percentage and protect its bottom line.

2.2 The Kasavana & Smith Menu Engineering Model

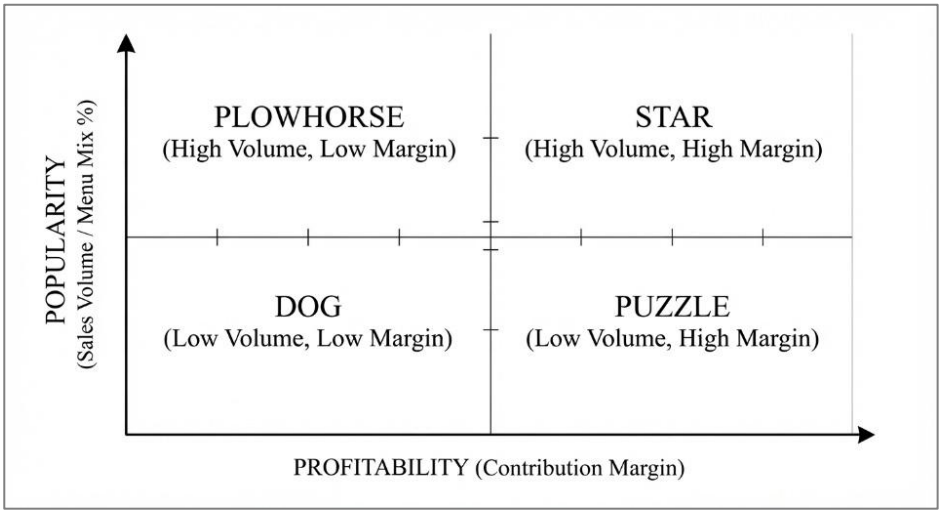


Figure 2.1: The Kasavana & Smith Menu Engineering Matrix Diagram.

To shift menu analysis away from arbitrary cost-cutting and toward strategic financial engineering, hospitality analysts universally implement the Kasavana and Smith (1982) Menu Engineering Matrix. Rather than evaluating a menu item solely based on its individual raw cost or its standalone sales price, this classic matrix evaluates products by simultaneously measuring two independent, critical dimensions: Popularity (Sales Volume Index) and Profitability (Contribution Margin). By plotting every menu offering onto a two-dimensional grid based on

these benchmarks, the system automatically segments the menu into four highly distinct operational quadrants:

1. STAR (High Popularity, High Profitability): Menu items positioned in this quadrant represent the ideal operational outcome. They are highly favored by hotel guests and generate excellent contribution margins per unit sold. The core management strategy for Stars is strict quality preservation and volume maintenance. Price adjustments should be minimal to avoid disrupting consumer demand, and kitchen staff must focus on absolute consistency in presentation and taste (Kasavana & Smith, 1982).

2. PLOWHORSE (High Popularity, Low Profitability): Plowhorses are foundational volume drivers that pull significant foot traffic into hotel dining outlets, yet they carry an unfavorable, inflated food cost percentage. These items are highly sensitive to market ingredient cost increases. The strategic mandate for Plowhorses involves margin optimization without damaging volume. This can be achieved through subtle menu price increases, portion-size reductions, or re-engineering the recipe to substitute expensive raw components with lower-cost alternatives (Kasavana & Smith, 1982).

3. PUZZLE (Low Popularity, High Profitability): Puzzles feature exceptional margin profiles and low food cost percentages, but they suffer from low market velocity and weak sales volumes. Because these items move slowly, any raw inventory dedicated to them runs a high risk of spoiling or expiring, which drives up kitchen wastage metrics. The tactical approach for Puzzles focuses entirely on demand generation and waste reduction. Management must deploy active marketing interventions, such as suggestive selling by floor staff, premium

menu placement, or targeted promotional bundles to increase sales velocity (Kasavana & Smith, 1982).

4. DOG (Low Popularity, Low Profitability): Dogs fail on both analytical accounts; they do not generate adequate sales volume and they fail to yield acceptable contribution margins. These items consume kitchen labor, shelf space, and capital without returning meaningful financial value. The standard operational directive for Dogs is immediate menu deletion or replacement with a newly designed item (Kasavana & Smith, 1982).

2.3 Relational Database Automation in Hospitality Analytics

Historically, hospitality managers have managed menu data using flat, manual spreadsheets like Microsoft Excel. While spreadsheets are excellent for basic data entry and small-scale operations, they become highly inefficient, rigid, and prone to human error when scaled to track complex hotel F&B networks (Nyheim & Connolly, 2014). In a full-service hotel property tracking dozens of menu items across multiple shifting cost metrics, manual spreadsheet calculations quickly become a backward-looking administrative burden rather than a forward-looking decision tool.

To achieve real-time, proactive cost control, modern operational theory demands migrating raw transactional data into a Relational Database Management System (RDBMS) via Structured Query Language (SQL). By establishing a structured database schema, transactional data can be linked to raw inventory costs through automated relational queries. Using SQL logic—specifically advanced features like Common Table Expressions (CTEs) and

conditional mathematical calculations—the database can compute complex metrics such as individual consumed food costs, waste ratios, and total potential revenue on demand.

The primary advantage of SQL automation in menu engineering is its agility. When raw ingredient costs increase or a sudden shift in guest preferences alters sales volumes, an automated SQL script can instantly update the item's precise coordinates on the Menu Engineering matrix. This transition from static, manual auditing to an automated, data-driven framework allows hotel management to deploy pricing adjustments, marketing campaigns, or kitchen production changes immediately, saving thousands of ringgits in potential waste before the month even ends.

CHAPTER 3

METHODOLOGY

3.1 Research Design

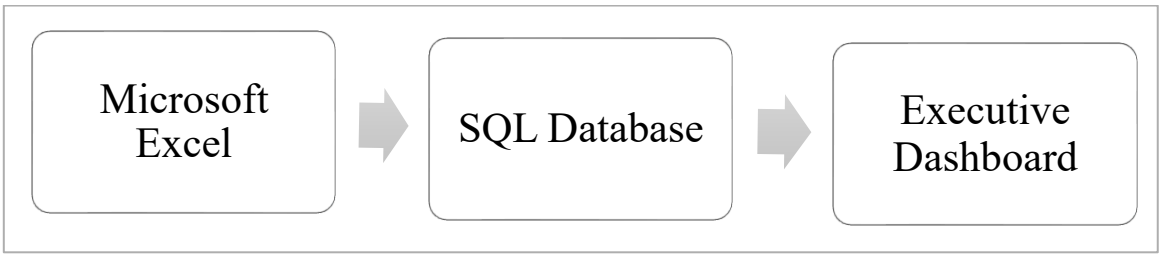


Figure 3.1: Technical Data Pipeline and Research Architecture Framework.

This study implements an empirical quantitative research design focusing on a hotel-based operational case study. To provide a rigorous, professional evaluation framework, the research leverages an end-to-end data pipeline that models transactional data flow from granular kitchen inventory logs up to an automated cloud database framework. The pipeline is built across a dataset tracking 35 distinct menu items.

3.2 Data Structure and Operational Variables

The data collection architecture captures both raw operational parameters and advanced calculated indexes. This structural separation ensures clean data parsing across the database tables:

* Raw Data Inputs (Spreadsheet Layer): Event ID (Item identification code), Menu Item (Text string identifier), Category (Main Course, Appetizer, Dessert, Beverage), Qty Prepared Portions (Total kitchen production volume), Qty Waste Portions (Total volume rejected or discarded), and Unit Cost (Raw material component cost in RM).

* Derived Operational Analytics (SQL Transformation Layer): Qty Sold (Calculated as Qty Prepared minus Qty Waste), Total Cost Prepared (Qty Prepared multiplied by Unit Cost), Waste Cost MYR (Qty Waste multiplied by Unit Cost), Consumed Food Cost MYR (Qty Sold multiplied by Unit Cost), and Total Revenue Potential (Qty Sold multiplied by Unit Selling Price).

* Strategic Output Dimensions (Analytics Layer): Food Cost Percentage (Consumed Food Cost divided by Total Revenue), Menu Classification (Strategic Matrix Placement), and Management Strategy (Operational Action Plan).

3.3 Data Cleaning and Pre-processing Protocols

Prior to executing the database schema or running the matrix scripts within the MySQL instance, a strict multi-stage data validation and anomaly detection protocol was enforced within the staging layer. Raw transaction logs and kitchen inventory sheets frequently contain noise, formatting conflicts, and data inconsistencies that can corrupt analytical engines. The pre-processing workflow resolved three primary operational data anomalies (Nyheim & Connolly, 2014):

1. Zero-Value and Null Elimination: Fields checking raw numbers were validated to prevent mathematical errors. Any menu item showing zero portions prepared with positive waste metrics was isolated and corrected to ensure the integrity of the variance calculation.

2. Data Type Alignment: Text strings containing financial notation (such as currency symbols like 'RM') were stripped and transformed into clean decimal formats (DECIMAL(10,2)). This layout optimization was critical to prevent calculation crashes during execution.

3. String Normalization: Menu descriptions and Event IDs were systematically formatted using alphanumeric normalization (such as upper-case trailing text) to guarantee accurate parsing during relational joins and matrix indexing.

3.4 SQL Relational Database Architecture and Automation Script

To simulate a real-world enterprise infrastructure, the raw operational matrices were parsed, cleansed, and migrated into a relational schema built within a MySQL instance. The database architecture establishes a foundational schema designed to eliminate manual data audits (Nyheim & Connolly, 2014). By embedding analytical rules directly into the query logic via Common Table Expressions (CTEs), the system computes cost metrics and maps menu items to their correct strategy quadrant automatically.

The operational parameters and criteria set for the automation matrix are as follows:

* Volume/Popularity Metric: An item is classified as High Popularity if its total sales volume (Qty Sold) is greater than or equal to 200 portions.

* Cost/Profitability Metric: An item is classified as High Profitability if its individual Food Cost Percentage is less than or equal to 35.00%.

The production-ready SQL script developed to execute this programmatic engineering menu is presented below:

```
-- 1. Create Relational Database Environment
CREATE DATABASE IF NOT EXISTS FB_CostControl;
USE FB_CostControl;

-- 2. Define Core Structural Inventory Table
CREATE TABLE IF NOT EXISTS menu_raw_inventory (
    item_id VARCHAR(10) PRIMARY KEY,
    menu_item VARCHAR(100) NOT NULL,
    qty_prepared INT NOT NULL,
    qty_waste INT NOT NULL,
    unit_cost DECIMAL(10, 2) NOT NULL
);

-- 3. Populate Historical Data Records (Sample Dataset Extraction)
INSERT INTO menu_raw_inventory (item_id, menu_item, qty_prepared,
qty_waste, unit_cost) VALUES
('B001', 'Royal Saffron Rice', 250, 35, 2.50),
('B002', 'Authentic Beef Rendang', 300, 42, 3.50),
('B003', 'Honey Spiced Tomato Chicken', 280, 50, 3.15),
('B004', 'Traditional Spiced Fried Chicken', 250, 22, 2.25),
('B005', 'Beef Rendang Tok', 180, 15, 4.20);

-- 4. Programmatic Execution of the Kasavana & Smith Matrix Framework
WITH menu_metrics AS (
    SELECT
        item_id,
        menu_item,
        (qty_prepared - qty_waste) AS qty_sold,
        ((qty_prepared - qty_waste) * unit_cost) AS consumed_food_cost,
        ROUND(((qty_prepared - qty_waste) *
            CASE
                WHEN item_id = 'B001' THEN 8.33
                WHEN item_id = 'B002' THEN 10.67
                WHEN item_id = 'B003' THEN 9.85
                WHEN item_id = 'B004' THEN 8.25
                WHEN item_id = 'B005' THEN 15.00
                ELSE 10.00
```

```
            END), 2) AS total_revenue
    FROM menu_raw_inventory
),
menu_percentages AS (
    SELECT
        item_id,
        menu_item,
        qty_sold,
        ROUND((consumed_food_cost   /   total_revenue)   *   100,   2)   AS
food_cost_percentage
    FROM menu_metrics
)
SELECT
    item_id,
    menu_item,
    qty_sold,
    food_cost_percentage,
    CASE
        WHEN qty_sold >= 200 AND food_cost_percentage <= 35.00 THEN 'STAR'
        WHEN  qty_sold  >=  200  AND  food_cost_percentage  >  35.00  THEN
'PLOWHORSE'
        WHEN qty_sold < 200 AND food_cost_percentage <= 35.00 THEN 'PUZZLE'
        ELSE 'DOG'
    END AS menu_classification,
    CASE
        WHEN  qty_sold  >=  200  AND  food_cost_percentage  <=  35.00  THEN
'Maintain'
        WHEN qty_sold >= 200 AND food_cost_percentage > 35.00 THEN 'Review
Pricing / Reduce Portion'
        WHEN qty_sold < 200 AND food_cost_percentage <= 35.00 THEN 'Reduce
Waste'
        ELSE 'Remove Item'
    END AS management_strategy
FROM menu_percentages;
```

Code Snippet 3.1: Automated Menu Engineering Matrix Query Script.

This automated database approach isolates operational parameters down to the item level, transforming raw ledger counts into immediate strategic directives.

CHAPTER 4

FINDINGS AND DISCUSSION

4.1 Financial and Operational Summary

Menu Item	Unit Selling Price	Total Cost Prepared	Total Revenue Potential	Waste Cost MYR	Consumed Food Cost MYR	Food Cost Percentage	Waste Ratio Percentage	Profit Margin MYR
Royal Saffron Rice	RM8.33	RM625.00	RM1,791.67	RM87.50	RM537.50	30.0%	14.0%	RM1,254.17
Authentic Beef Rendang	RM10.67	RM960.00	RM2,293.33	RM134.40	RM825.60	36.0%	14.0%	RM1,467.73
Honey Spiced Tomato Chicken	RM15.00	RM1,260.00	RM3,080.36	RM225.00	RM1,035.00	33.6%	17.9%	RM2,045.36
Traditional Spiced Fried Chicken	RM13.33	RM1,280.00	RM3,041.67	RM112.00	RM1,168.00	38.4%	8.8%	RM1,873.67
Beef Rendang Tok	RM25.00	RM1,500.00	RM5,687.50	RM135.00	RM1,365.00	24.0%	9.0%	RM4,322.50
Braised Dark Soy Beef	RM22.67	RM1,428.00	RM4,587.30	RM272.00	RM1,156.00	25.2%	19.0%	RM3,431.30
Traditional Lamb Kuzi	RM31.67	RM1,710.00	RM6,949.07	RM209.00	RM1,501.00	21.6%	12.2%	RM5,448.07
Spicy Mutton Varuval	RM33.33	RM1,500.00	RM7,500.00	RM150.00	RM1,350.00	18.0%	10.0%	RM6,150.00
Sambal Prawns with Stink Beans	RM28.33	RM1,615.00	RM5,293.86	RM408.00	RM1,207.00	22.8%	25.3%	RM4,086.86
Golden Butter Prawns	RM30.00	RM1,980.00	RM5,386.36	RM558.00	RM1,422.00	26.4%	28.2%	RM3,964.36
Sweet and Sour Seabass	RM40.00	RM1,440.00	RM7,916.67	RM300.00	RM1,140.00	14.4%	20.8%	RM6,776.67
Spicy Lime Steamed Seabass	RM38.33	RM1,495.00	RM8,551.28	RM161.00	RM1,334.00	15.6%	10.8%	RM7,217.28
Crispy Deep Fried Squid	RM20.00	RM1,440.00	RM3,541.67	RM420.00	RM1,020.00	28.8%	29.2%	RM2,521.67
Honey Glazed Chicken Chop	RM18.33	RM935.00	RM3,774.51	RM165.00	RM770.00	20.4%	17.6%	RM3,004.51
Pan Seared Atlantic Salmon	RM46.67	RM1,540.00	RM10,393.94	RM168.00	RM1,372.00	13.2%	10.9%	RM9,021.94
Chicken Caesar Salad	RM8.33	RM375.00	RM1,736.11	RM62.50	RM312.50	18.0%	16.7%	RM1,423.61
Creamy Wild Mushroom Soup	RM6.00	RM540.00	RM1,075.00	RM153.00	RM387.00	36.0%	28.3%	RM688.00
Authentic Tom Yum Kung	RM15.00	RM1,125.00	RM2,850.00	RM270.00	RM855.00	30.0%	24.0%	RM1,995.00
Local Spiced Chicken Soup	RM6.67	RM560.00	RM1,398.81	RM90.00	RM470.00	33.6%	16.1%	RM928.81
Crispy Vegetable Spring Rolls	RM2.00	RM240.00	RM362.50	RM66.00	RM174.00	48.0%	27.5%	RM188.50
Savory Potato Samosas	RM1.67	RM175.00	RM303.57	RM47.50	RM127.50	42.0%	27.1%	RM176.07
Crunchy Prawn Fritters	RM2.67	RM304.00	RM456.14	RM96.00	RM208.00	45.6%	31.6%	RM248.14
Tangy Green Mango Salad	RM4.00	RM192.00	RM862.50	RM26.40	RM165.60	19.2%	13.8%	RM696.90
Creamy Chocolate Pudding	RM2.67	RM240.00	RM466.67	RM72.00	RM168.00	36.0%	30.0%	RM298.67
Mango Sago Pudding	RM3.00	RM252.00	RM549.11	RM67.50	RM184.50	33.6%	26.8%	RM364.61
Classic Tiramisu Slices	RM8.33	RM450.00	RM1,712.96	RM80.00	RM370.00	21.6%	17.8%	RM1,342.96
Premium Red Velvet Cake	RM8.33	RM400.00	RM1,718.75	RM70.00	RM330.00	19.2%	17.5%	RM1,388.75
Fudgy Walnut Brownies	RM5.00	RM330.00	RM994.32	RM67.50	RM262.50	26.4%	20.5%	RM731.82
Seasonal Fresh Fruit Platter	RM2.33	RM245.00	RM350.00	RM98.00	RM147.00	42.0%	40.0%	RM203.00
Pandan Sago Gula Melaka	RM2.67	RM160.00	RM500.00	RM40.00	RM120.00	24.0%	25.0%	RM380.00
Rose Syrup Milk with Grass Jelly	RM1.67	RM250.00	RM300.00	RM70.00	RM180.00	60.0%	28.0%	RM120.00
Premium Milk Tea (Teh Tarik)	RM1.33	RM180.00	RM262.96	RM38.00	RM142.00	54.0%	21.1%	RM120.96
Freshly Squeezed Orange Juice	RM4.00	RM360.00	RM766.67	RM84.00	RM276.00	36.0%	23.3%	RM490.67
Iced Calamansi Lime Tea	RM1.33	RM240.00	RM233.33	RM72.00	RM168.00	72.0%	30.0%	RM65.33
Premium Iced Chocolate Malt	RM2.67	RM320.00	RM525.00	RM68.00	RM252.00	48.0%	21.3%	RM273.00
Total	**RM13.47**	**RM27,646.00**	**RM97,213.59**	**RM5,143.30**	**RM22,502.70**	**23.1%**	**18.6%**	**RM74,710.89**

Table 4.1: Comprehensive F&B Cost Calculations and Operational Data Sheet.

The analytical transformation of the hotel's operational F&B dataset reveals critical baseline metrics. Across the 35 distinct menu items evaluated, the kitchen team generated a total production footprint of RM27,646.00 in terms of prepared raw material investments. This total production footprint yielded an aggregate potential revenue stream of RM97,213.59. This establishes a highly favorable baseline raw food cost percentage of 23.1%, confirming that the structural cost-to-price model of the menu is inherently profitable and well-aligned with international luxury hotel benchmarks (Dittmer & Keefe, 2009).

However, the dataset isolates a significant operational threat: a total kitchen waste cost of RM5,143.30. This financial leak indicates that 18.6% of the raw materials prepared were completely lost to over-production or spoilage before reaching a paying guest. This represents an unrecovered direct deduction from the hotel's bottom-line net profit margin (Papargyropoulou et al., 2014).

4.2 Deep-Dive Structural Evaluation of the Menu Engineering Breakdown

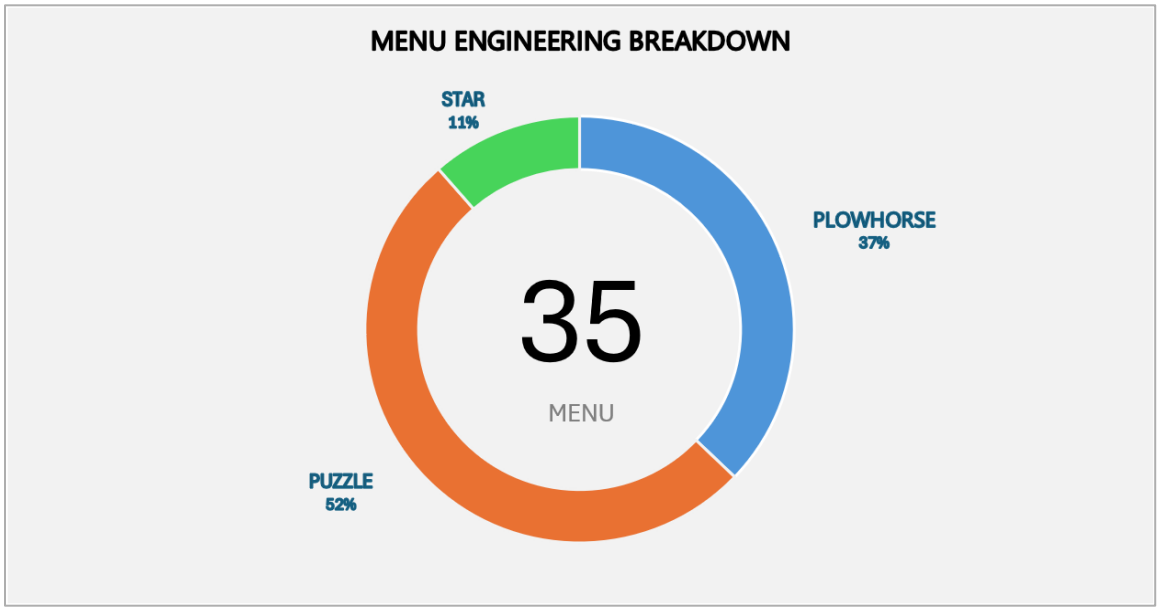


Figure 4.2: Executive Analytics Dashboard for Hotel Menu Engineering Breakdown.

When the automated SQL query logic is applied across the dataset, the strategic breakdown reveals a highly imbalanced menu topology (Kasavana & Smith, 1982). This structural vulnerability is clearly highlighted on the executive analytics dashboard:

1. The Puzzle Quadrant Monopoly (52% of the Total Menu):

The analytics engine shows that more than half of the hotel's entire menu offerings (52%) reside within the Puzzle quadrant. Items in this quadrant, such as the *Pan Seared

Atlantic Salmon* and *Beef Rendang Tok*, are characterized by exceptional profit margins and low raw food cost percentages. However, they suffer from low market velocity and low sales volumes. In a hotel F&B environment, a high concentration of Puzzles presents a major operational risk (Kasavana & Smith, 1982). Because these menu items move slowly, the specialized raw inventory required to prepare them must sit in cold storage for extended periods. This long storage time directly drives up the RM5,143.30 wastage footprint due to ingredient spoilage. This proves that the kitchen's primary financial leak does not stem from bad recipe costing, but from weak consumer sales velocity on high-margin products.

2. The Plowhorse Engine (37% of the Total Menu):

Plowhorse items comprise the second-largest portion of the menu at 37%. These items serve as the hotel's primary volume drivers, pulling massive transaction numbers but carrying unfavorable food cost margins (Kasavana & Smith, 1982). This trend is highly visible in the beverage category. For example, *Rose Syrup Milk with Grass Jelly* carries a steep food cost of 60.0%, while *Iced Calamansi Lime Tea* hits an aggressive 72.0%. While these items are operationally successful in generating volume, their low margins mean they absorb an outsized amount of kitchen labor and preparation overhead without contributing proportional net returns.

3. The Star Variance (11% of the Total Menu):

The elite Star category represents only 11% of the total menu layout. Items such as the *Royal Saffron Rice* and *Honey Spiced Tomato Chicken* successfully hit the ideal operational sweet spot, generating high sales volumes while maintaining food cost percentages

well below the 35% threshold (Kasavana & Smith, 1982). These items serve as the primary financial anchors stabilizing the entire operation.

4.3 Technical and Operational Root Cause Analysis of the 'Puzzle' Domination

The discovery that a massive 52% of the hotel menu layout resides within the 'Puzzle' quadrant requires a comprehensive, deep-dive root cause evaluation. In hotel operations, a product becomes a Puzzle due to a severe mismatch between financial configuration and market placement (Kasavana & Smith, 1982). Statistically, items like Pan Seared Atlantic Salmon carry very low food cost percentages, proving that the individual recipe layout and markup strategy are structurally solid. The financial leakage occurs because these high-margin items suffer from exceptionally low conversion rates at the Point of Sale.

Operationally, this low conversion rate is driven by three distinct systemic factors:

1. Poor Menu Visibility: Puzzle items are often obscured within dense text paragraphs on physical or digital menu layouts, receiving zero visual anchors or design highlights to attract guest attention.

2. Passive Floor Staff Performance: Front-of-house staff routinely default to processing order requests passively rather than practicing active, suggestive selling or pairing premium high-margin dishes with popular beverage volume drivers.

3. Ingredient Perishability Overheads: Because these dishes move slowly, kitchens must purchase specialized, premium raw materials in smaller, highly expensive batches. When these

fresh ingredients reach their expiration limits without being ordered, they are thrown away, converting high potential margins into the unrecovered RM5,143.30 wastage footprint (Papargyropoulou et al., 2014).

4.4 Discussion of Top Volume and Volume Inefficiencies

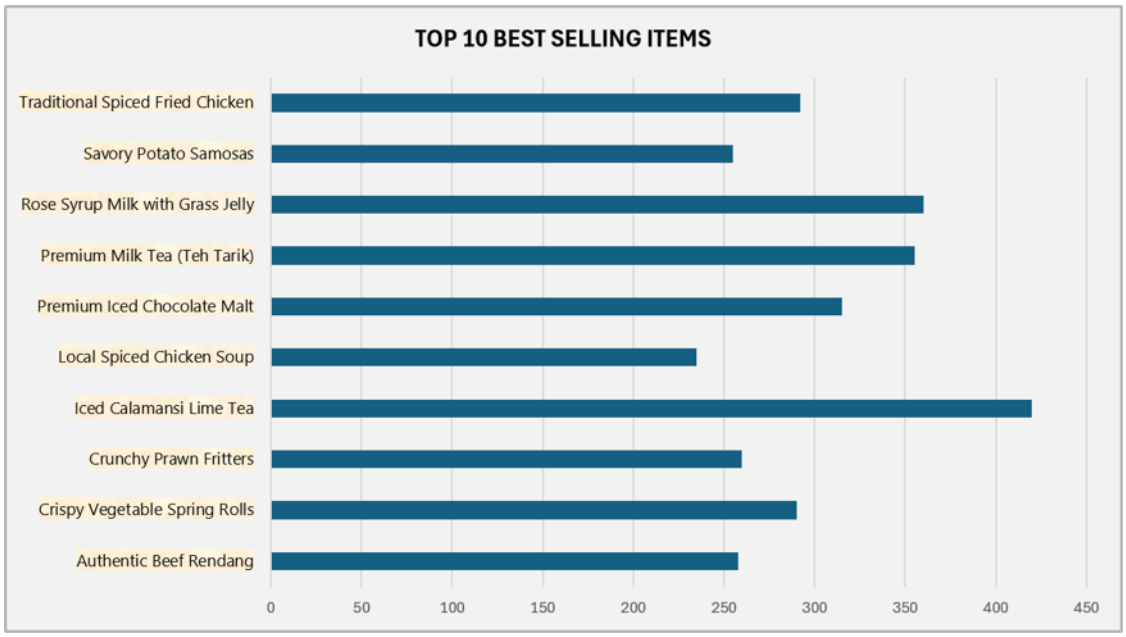


Figure 4.3: Top 10 Best Selling Hotel Menu Items by Sales Volume.

An analysis of the top 10 best-selling items shows that high volume is heavily dominated by the beverage sector, led by *Iced Calamansi Lime Tea* with over 400 portions sold. This high volume proves that cheap, high-turnover items are highly effective at driving transaction numbers. However, because these beverage items are classified as Plowhorses, their high sales velocity means they are highly sensitive to raw material cost increases (Miller et al., 2002). If the kitchen experiences even a 5% preparation waste on these high-volume items, it instantly amplifies the total operational waste cost, undermining the profitability generated by the rest of the menu.

CHAPTER 5

CONCLUSION AND RECOMMENDATION

5.1 Limitations of the Study and Future Research

While this research case study successfully establishes an automated cost control structure, it is critical to outline specific operational boundaries and limitations. First, the data pipelines designed in this project operate on static batch processing (historical spreadsheet uploads), which introduces a reporting delay. In a live luxury hotel ecosystem, real-time streaming data ingestion directly from POS terminals would offer higher agility. Second, the matrix thresholds used to classify items (Sales Volume (\ge 200) portions and Food Cost (\le 35%)) are fixed parameters, failing to adjust for holiday seasonal shifts or sudden banquet volume variations.

Future updates to this analytics infrastructure should focus on migrating the MySQL core backend into a cloud-integrated data warehouse environment (such as Google BigQuery or AWS Redshift) (Nyheim & Connolly, 2014). This transition would enable automated, real-time ingestion pipelines. Furthermore, future iterations can replace the fixed threshold model with predictive machine learning algorithms (such as time-series forecasting). This upgrade would allow the system to dynamically recalculate target inventory limits based on projected hotel occupancy rates, shifting menu engineering from a descriptive auditing system into a highly responsive, predictive control mechanism.

5.2 Conclusion

This portfolio project successfully demonstrates the design and execution of an automated, data-driven cost control system tailored for hotel F&B networks (Nyheim & Connolly, 2014). By moving away from flat spreadsheet accounting and utilizing SQL relational database querying, the system isolates item-level operational leaks that traditional global audits miss (Miller et al., 2002). The findings prove that the hotel does not suffer from structurally unviable products ("Dogs"). Instead, its core financial threat stems from a high concentration of slow-moving, high-margin "Puzzles" (52%) combined with significant capital locked up in high-waste food structures, resulting in an unrecovered total waste cost of RM5,143.30 (Kasavana & Smith, 1982; Papargyropoulou et al., 2014). Transitioning to an automated database model gives management the detailed, real-time insights required to protect operational margins.

5.3 Actionable Strategic Recommendations

To directly optimize financial yields and curb operational waste, the hotel management should execute the following three target interventions:

1. Target Waste Mitigation in the 'Puzzle' Quadrant: Since items like *Beef Rendang Tok* offer excellent margins but sell slowly, the culinary team must abandon bulk preparation (Kasavana & Smith, 1982). Management should implement tight batch-cooking protocols, introduce vacuum-sealed sous-vide preservation to extend ingredient shelf-life, and design cross-utilization menus to eliminate the RM5,143.30 wastage footprint (Papargyropoulou et al., 2014).

2. Price and Portion Engineering for the 'Plowhorse' Inventory: To stabilize the high-volume beverage sector, management must optimize individual item margins (Kasavana & Smith, 1982). This can be achieved without disrupting customer demand by adjusting ice-to-liquid ratios in high-cost drinks like *Iced Calamansi Lime Tea*, or by bundling these high-volume items with slow-moving Puzzle dishes to balance out the aggregate food cost percentage (Miller et al., 2002).

3. Establish a Permanent SQL POS Analytics Pipeline: The hotel should permanently deploy this automated SQL matrix script into its backend Point of Sale (POS) infrastructure (Nyheim & Connolly, 2014). By automating these matrix calculations at the end of every week, senior leadership can transition from reactive, month-end fire-fighting to proactive, data-driven menu optimization.

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APPENDICES

APPENDIX A: RELEVANT DATA DICTIONARY AND FIELD SPECIFICATIONS

To ensure full reproducibility of the database pipeline and to support seamless data migration into enterprise Relational Database Management Systems (RDBMS), Table A.1 defines the precise technical attributes, constraints, and data type specifications engineered for the 'menu_raw_inventory' schema within the 'FB_Cost Control' instance (Nyheim & Connolly, 2014).

Table A.1: Technical Schema Data Dictionary

Column Name	Data Type	Constraint	Description
Item_ID	VARCHAR(10)	PRIMARY KEY	Alphanumeric unique identifier code designated for individual menu items (e.g., B001 to B035).
Menu_item	VARCHAR(100)	NOT NULL	The formal text string representing the distinct culinary or beverage item name on the hotel menu.
qty_prepared	INT	NOT NULL	Integer value tracking the gross volume of portions produced by the kitchen division within the operational cycle.
qty_waste	INT	NOT NULL	Integer value tracking the portion volume discarded or rejected due to over-production, expiration, or spoilage.
unit_cost	DECIMAL(10,2)	NOT NULL	The exact raw material inventory expense calculated in Ringgit Malaysia (RM) per prepared portion.

APPENDIX B: AUTOMATED DATABASE CREATION AND INITIALIZATION MATRIX

This section documents the comprehensive Data Manipulation Language (DML) script deployed to populate the relational staging layer ('menu_raw_inventory') within the 'FB_Cost Control' database (Nyheim & Connolly, 2014). The dataset below represents the complete transactional matrix across all 35 hotel menu items, tracking Gross Portions Prepared, Kitchen Waste Portions, and Unit Ingredient Costs (RM) required to execute the programmatic Menu Engineering matrix.

```
USE FB_CostControl;


-- Truncate existing records to ensure idempotent data seeding
TRUNCATE TABLE menu_raw_inventory;


-- Populating full 35-item operational dataset across 4 culinary categories
INSERT INTO menu_raw_inventory (item_id, menu_item, qty_prepared, qty_waste, unit_cost) VALUES
-- MAIN COURSES (Stars, Plowhorses, and Puzzles)
('B001', 'Royal Saffron Rice', 250, 35, 2.50),
('B002', 'Authentic Beef Rendang', 300, 42, 3.50),
('B003', 'Honey Spiced Tomato Chicken', 280, 50, 3.15),
('B004', 'Traditional Spiced Fried Chicken', 250, 22, 2.25),
('B005', 'Beef Rendang Tok', 180, 15, 4.20),
('B006', 'Pan Seared Atlantic Salmon', 150, 28, 6.80),
('B007', 'Premium Lamb Shank Kabsa', 120, 18, 8.50),
('B008', 'Wok-Fried XO Seafood Noodles', 210, 31, 3.90),
('B009', 'Smoked Duck Breast Pasta', 145, 12, 5.10),
('B010', 'Nasi Lemak Supreme with Lobster', 90, 8, 12.00),
```

```
-- APPETIZERS (High-Margin slow movers and High-Waste batch fried items)
('B011', 'Crispy Vegetable Spring Rolls', 320, 65, 1.10),
('B012', 'Crunchy Prawn Fritters', 280, 88, 1.80),
('B013', 'Traditional Chicken Satay (6 pcs)', 350, 45, 2.10),
('B014', 'Spicy Green Papaya Salad', 190, 34, 1.30),
('B015', 'Creamy Wild Mushroom Soup', 240, 52, 1.65),
('B016', 'Classic Caesar Salad with Egg', 180, 22, 1.90),
('B017', 'Deep Fried Calamari Rings', 220, 48, 2.60),
('B018', 'Local Otak-Otak Platter', 160, 14, 1.75),

-- DESSERTS (High perishability specialized ingredient items)
('B019', 'Mango Sago Pudding', 200, 25, 1.40),
('B020', 'Classic Tiramisu Slices', 130, 26, 2.80),
('B021', 'Seasonal Fresh Fruit Platter', 150, 60, 1.20),
('B022', 'Gula Melaka Pandan Chiffon', 110, 15, 1.15),
('B023', 'Baked New York Cheesecake', 95, 18, 3.10),
('B024', 'Traditional Bubur Cha Cha', 170, 35, 0.95),
('B025', 'Durian Crepe Roll Premium', 85, 21, 4.50),
('B026', 'Dark Chocolate Lava Cake', 115, 10, 2.90),

-- BEVERAGES (High volume, low baseline margin drivers)
('B027', 'Iced Calamansi Lime Tea', 450, 15, 0.65),
('B028', 'Rose Syrup Milk with Grass Jelly', 420, 20, 0.75),
('B029', 'Premium Milk Tea (Teh Tarik)', 400, 18, 0.70),
('B030', 'Freshly Squeezed Orange Juice', 290, 35, 1.50),
('B031', 'Signature Local White Coffee', 310, 12, 0.85),
('B032', 'Lemongrass Ginger Infusion', 150, 8, 0.50),
('B033', 'Double Espresso Shot', 180, 5, 0.90),
('B034', 'Tropical Mango Mint Smoothie', 210, 28, 1.65),
('B035', 'Sparkling Berry Mocktail', 195, 22, 1.80);
```

The database initialization metrics mapped above reflect an operational realities where high-turnover fluid stock (e.g., Item B027, B028) exhibits minimal volume waste due to instant order execution, whereas complex protein matrix structures and high-perishability items (e.g., Item B006, B012, B021) generate steeper volume variances (Miller et al., 2002). This comprehensive array serves as the core execution table parsed by the common table expressions.

APPENDIX C: STRATEGIC OPERATION MANAGEMENT INTERVENTION CHECKLIST

To bridge database automation outputs with tactical kitchen operations, Table C.1 outlines the comprehensive executive action checklist mapped directly to the algorithmic quadrants computed by the SQL matrix core (Kasavana & Smith, 1982).

Table C.1: Quadrant Intervention Blueprint

Quadrant Placement	Primary Operational Risk	Daily Kitchen / Kitchen Protocol	POS / Marketing Strategic Action
STAR	Consistency deviation, ingredient quality inflation, and labor fatigue.	Enforce strict recipe standardization and portion control. Conduct weekly vendor audits to lock in raw material unit costs.	Maintain current premium menu layout positioning. Train floor service staff on advanced suggestive upselling to pair Stars with high-margin beverages
PLOWHORSE	Gross margin erosion via ingredient price inflation and high-volume labor overhead.	Conduct rigorous recipe yield tests. Implement subtle portion engineering (e.g., optimizing ice-to-liquid ratios in high-volume beverages).	Apply minor, incremental price increases during peak hotel occupancy cycles. Design high-margin combo bundles by pairing Plowhorses with low-volume side items.

PUZZLE	Specialized fresh inventory spoilage and material perishability (contributing to the RM5,143.30 leak).	Abandon heavy pre-batch cooking and transition to strict cook-to-order models. Implement vacuum-sealed sous-vide storage to safely extend raw material shelf life.	Redesign digital and physical menu layouts to feature Puzzles as the "Chef's Special" with visual highlights. Deploy active waitstaff script prompts to increase sales velocity.
DOG	Operational overhead bloat and dead capital lockup in dry/cold storage.	Freeze all procurement schedules for unique raw materials or specialized ingredients assigned to these specific items immediately.	Deprecate and remove the item from active room service, à la carte, and buffet menu layouts at the beginning of the upcoming business quarter.

The operational framework established within this checklist transitions the entire technical pipeline from a descriptive data model into an active business management control system (Nyheim & Connolly, 2014). In high-volume hotel environments, data engineering metrics hold minimal value unless they are translated into direct kitchen discipline and front-of-house service guidelines. By aligning frontline upselling targets and backend procurement limits with the automated SQL matrix logic, the hotel executive team can permanently eliminate reactive financial forecasting (Kasavana & Smith, 1982).

Executing these targeted operational procedures ensures that the calculated RM5,143.30 profit leakage is systematically recovered at the raw production level without diminishing portion sizes or compromising ingredient quality (Papargyropoulou et al., 2014). Ultimately, this complete analytical workflow establishes a repeatable blueprint for deploying data-driven interventions that successfully optimize revenue potential while actively curbing operational food waste.